

JIYA YADAV

AI/ML Engineer

[LinkedIn](#) | jiyayadav102@gmail.com | [GitHub](#) | +977 9814735259 | Kathmandu, Nepal | [Portfolio](#) | [Hugging face](#)

PROFESSIONAL SUMMARY

AI/ML Engineer building end-to-end production ML systems. Fine-tuned transformer models achieving 0.89 macro F1 with 96% recall on crisis-signal detection, architected multi-agent async orchestration reducing inference latency by 40%, and established production CI/CD pipelines with 74% test coverage. MSc Information Technology (AI/ML) — Leeds Beckett University (completed, results expected Aug 2025). Focused on transparent, reliable, and production-ready AI systems.

TECHNICAL SKILLS

Machine Learning: PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost, SHAP, Fairlearn, Fine-tuning, RAG

Programming: Python (advanced), SQL, AsyncIO

ML Engineering: FastAPI, Streamlit, Pydantic v2, ChromaDB, sentence-transformers, Docker

MLOps & DevOps: Git, GitHub Actions (CI/CD), pytest, Codecov, Apache Airflow, Hugging Face Hub, Ruff

Data & Analytics: Pandas, NumPy, PostgreSQL, MySQL, Power BI

Cloud & Deployment: Hugging Face Spaces, Google Colab

FEATURED PROJECT

Mental Health Agentic AI Platform

Live Demo: huggingface.co/spaces/YDVJIYA/mental-health-ai-platform

Source: github.com/JIYA-YDV/Mental-Health-Agentic-AI-Platform

Model: huggingface.co/YDVJIYA/distilroberta-base-finetuned-emotion

Production multi-agent NLP platform for emotion classification and crisis detection with async orchestration, retrieval-augmented generation, and full MLOps tooling.

- Fine-tuned DistilRoBERTa on 16K labeled emotion samples using AdamW + linear warmup on Google Colab T4 GPU, improving baseline macro F1 from 0.87 to 0.89 with 96% recall on crisis signals. Published model to Hugging Face Hub with reproducible training notebook.
- Architected async multi-agent orchestration (Classification → Crisis + RAG in parallel → Aggregation) using asyncio.gather, reducing latency by 40% versus sequential execution.
- Established production test suite (69 tests, 74% coverage) with mocked ML dependencies enabling sub-8-second CI feedback via GitHub Actions matrix testing on Python 3.10.
- Deployed to Hugging Face Spaces (16GB free tier) with UptimeRobot monitoring; implemented a layered safety-override system protecting against domain-shift misclassifications on clinical vocabulary.

Stack: PyTorch • Transformers • FastAPI • Streamlit • ChromaDB • pytest

ADDITIONAL PROJECTS

Fair & Explainable ML for Mental Health Treatment Prediction | MSc Dissertation

- Built an interpretable ML pipeline predicting mental health treatment outcomes on 4,600 samples from the OSMI Mental Health in Tech Survey (8 waves, 2014–2022), using XGBoost + Explainable Boosting Machines.
- Applied a four-method explainability stack — SHAP (global + local), LIME (instance-level), Neural Additive Models (NAM), and EBM — to compare attribution consistency across model families and conducted algorithmic fairness analysis using Fairlearn across diverse demographic subgroups; implemented rigorous data-leakage testing to validate temporal train/test splits.

Stack: Python • Scikit-learn • XGBoost • EBM • NAM • SHAP • LIME • Fairlearn

E-Commerce Customer Lifetime Value Optimization | github.com/JIYA-YDV/E-Commerce-Customer-Lifetime-Value-Optimization

- Built RFM segmentation (6 tiers), identifying 26% of customers driving 62% revenue; performed cohort retention analysis and FP-Growth market basket analysis for cross-sell opportunities. End-to-end analytics on 393K+ transactions from a UK-based retailer (£7.2M annual revenue, 4.3K customers, 37 countries) — from raw data cleaning through PostgreSQL warehouse to Power BI dashboards.

Stack: Python • Pandas • Scikit-learn • MLxtend • PostgreSQL • SQLAlchemy • Power BI • DAX

Automated ETL Pipeline with Airflow & PostgreSQL | <https://github.com/JIYA-YDV/automated-etl-pipeline-airflow-postgresql>

- Production ETL ingesting hourly weather data for 5 global cities via OpenWeatherMap API into PostgreSQL, orchestrated with Apache Airflow. Implemented 25 automated data-quality checks per run and 27 pytest unit tests (100% passing); deployed on WSL2 + Ubuntu 22.04.

Stack: Python 3.10 • Apache Airflow 2.8 • PostgreSQL • pytest • WSL2

WORK EXPERIENCE

Backend AI Engineering Intern | Jul 2026 – Present | FlyRank AI

- Building backend Python services for an AI product at an early-stage startup, contributing to model integration and API design alongside senior engineers.

Digital Marketing & Research Analyst | May 2022 – May 2023 | GCG International LLC, Kathmandu

- Executed analytics-driven digital marketing campaigns with quantitative KPI tracking and consumer behavior analysis to inform data-driven business decisions.

EDUCATION

MSc Information Technology (AI/ML) | | **Expected result: Sep 2026** | Leeds Beckett University via The British College, Kathmandu

BSc (Hons) Computing — First Class Honours | 2021 – 2024 | Leeds Beckett University via The British College, Kathmandu

CERTIFICATIONS

- Anthropic — AI Fluency: Framework & Foundations
- Anthropic — Claude Code 101
- Hugging Face — Fundamentals of Agents
- Udemy — SQL for Data Analytics: Zero to Hero